William Marietta

Montclair, VA | wemarietta@gmail.com | wmarietta.github.io | github.com/wmarietta

EDUCATION

George Mason University — Bachelor of Science in Computer Science May 2026

GPA: 3.67/4.00

Relevant Coursework: Web App Development, Database Concepts, Secure Programming, Distributed Systems

PROJECTS

Duelist Kingdom — Full-Stack Web Application *React, TypeScript, Node.js, Express, MySQL*

- **Architected a full-stack application** with a React/TypeScript frontend and layered Node.js/Express REST API, separating request handling, business logic, and data access to improve maintainability and extensibility.
- **Designed a relational MySQL database** for a trading card game management platform, implementing deck-management APIs, YDK deck imports, and cached external card searches to reduce redundant requests.
- **Implemented secure authentication** using Argon2 password hashing, persistent sessions, and server-side session and ownership checks that restrict deck changes to their owners.

Battleframe — Native Android Turn-Based Fighting Game *Java, Android SDK, Gson*

- **Engineered a custom 2D animation system** using Android's SurfaceView and a dedicated rendering thread, separating frame updates and drawing from the main UI thread while using shared update/draw interfaces for characters and projectiles.
- **Designed an extensible turn-based combat system** with reusable interfaces for abilities and items, JSON-driven game content, and integrated character progression.

Distributed MapReduce — Distributed Systems *Go, Concurrency*

- **Developed a distributed MapReduce framework in Go**, coordinating concurrent worker processes to execute computational workloads in parallel.
- **Implemented task scheduling and state tracking** to enforce dependencies between map and reduce phases across workers.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Java, Go, Python, C++, C#
Frameworks & APIs	React, Node.js, Express, REST APIs, Android SDK
Databases	MySQL, MongoDB
Tools & Practices	Git, Unit Testing, Agile/Scrum, Android Studio, Unity, Unreal Engine 5

EXPERIENCE

Mason Game and Technology Academy — Lead Instructor 2016–2024

- Led programming and game-development classes of 20+ students through hands-on software projects.
- Developed technical curricula, programming exercises, and course materials for new and existing classes.
- Trained and mentored new instructors and teaching assistants on technical course content.

Virginia Serious Game Institute — Instructor 2017–2023

- Taught Unity application development to groups of 6–20+ public school teachers and professionals.
- Developed technical curricula and materials, and contributed to an interactive educational web platform.
- Presented technical demonstrations and educational software to prospective business partners.